Does inferred demonstrator expertise transfer to a new environment? Learning from demonstrations in multi-armed bandits

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Project links

- **Live experiment:** https://asteroid-mining-game-31128277324.us-east4.run.app/ (the version deployed for this study; take the study exactly as participants do)
- **Experiment code:** https://github.com/martinezjulio/AsteroidMiningAdenture, tag `[TODO: 6.1]`
- **Modelling and analysis code:** https://github.com/martinezjulio/SocialTransformer, tag `[TODO: 6.1]`
- **Transformer checkpoint:** attached to the registration; sha256 `9f3586bd30704526da7042c87f5492a6109867b5eb50dc8b4ddf211f538decac`
- **Preregistration:** OSF registration, link `[TODO: after posted to OSF]`; registered before recruitment

AI use statement

I used Claude Code (various versions of Sonnet, Opus 5.5 and Fable 5.1) to help me implement the experiment code, the modeling and analysis pipeline and tests. Claude Code was also used to set up this document, its Quarto structure, rendering and the tables of pipeline numbers. The text of this registration is my own with edits from Claude Fable 5.1.

1 Introduction

This study asks whether a social learner's theory of mind representations of a social partner, i.e. their latent environmental beliefs and experience, shape the learner's choices, and results in improved outcomes in a new environment with prior exposure to the same social partner. We study this in a multi-armed

bandit task where participants both play and watch demonstrations of simulated partners with different levels of experience.

2 Task and design

2.1 Materials

The task. The task is a 12-armed Bernoulli bandit themed as an asteroid mining adventure game. Participants mine by shooting asteroids, which represent the arms of the game; shooting one results in a reward of 0 or 1 based on a hidden underlying payoff probability. All bandits re-used the same set of payoff probabilities from $\{0, 0.1, 0.2, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.75, 0.8\}$ where different bandits are permutations of these payoffs.

Games. Conditions in the task are made up of games. Each game is either a play game or a watch game. Play games allow exactly 6 trials of play on the bandit done by the participant. Watch games also allow exactly 6 trials but they are done by a simulated partner which the participant watches. Both types of games show the outcomes of the trials, both the choice that was made and the reward outcome received. In addition there is a choice game that only applies to the social conditions. The choice game is a single trial where participants are asked to select one of the two demonstrator partners they watched during the watch games.

Partners. There are two types of demonstrator partners that can be watched during the watch games: an expert or a novice. Both are Thompson samplers where the novice has no prior experience and hence starts from uniform prior beliefs whereas the expert has 128 trials of prior experience that forms their beliefs over the bandit. Experience however is hidden to the participant and can only be inferred from observing their demonstrations and outcomes. One key detail participants are informed about is that partners that have more experience have more experience across all bandits and vice versa for novice partners. Thompson samplers for the demonstrators are generated prior to each watch game and only filters for the expert's choices being better in expectation than the novice's on that bandit; however, this doesn't prevent reward outcomes from favouring the novice.

2.2 Procedure

Order of events.

1. Consent
2. Instructions
3. Practice session
4. 3 Main conditions
5. Return screen

The practice session is a sequence of games defined above as [Play, Watch, Play] all on the same bandit for familiarization and not analyzed.

Conditions. The design is within-participants: there are 3 conditions and every participant completes all three, in random order. 2 are social conditions made up of watch and play games. The third is a non-social or solo condition made up entirely of play games. Each condition is split into rounds. Each round repeats the same set of games again but in a new bandit with new payoffs. In the social conditions both an expert and a novice demonstrator are watched in each round. The order of the type of demonstrator watched first is counterbalanced across participants. The social conditions end with a choice game whereas the solo condition does not. The table below shows the structure of each condition.

Condition	Each round	Choice game
expert-first	watch expert, **play**, watch novice, **play**	true
novice-first	watch novice, **play**, watch expert, **play**	true
solo	**play**, **play**, **play**, **play**	false

Outcomes. Three outcomes are measured in the play games: the mean of the true payoffs of the chosen arms, the mean of observed rewards, and the mean entropy of the arm choices (in bits).

Randomization. Condition order, payoff probabilities in each bandit (for each round), outcome rewards themselves, bandit background images in the game (corresponding to each bandit), asteroid images for each arm for each bandit, and the order partners are presented in choice games are all randomized.

3 Hypotheses

The list below shows each confirmatory claim, both primary and secondary. For each confirmatory claim the estimand, the model from which it is derived and the accompanying rule are listed. The primary claims use the H labels (i.e. H1 through H5) and the secondary claims use the S label (i.e. S1 through S4). The contrast expressions for the H1, H2, H3 and H5 claims are included; the models from which these contrasts are derived are specified in Section 4. In the contrast expressions, `emm <- emmeans(fit, ~ condition * play * round)` is a twelve-cell grid of the fitted descriptive model, `m(sel)` is the weight vector that averages the selected cells, and `demo` is the demonstrator watched before the play game (`"none"` in the solo condition), as in `src/r/descriptive.R`.

Primary Claims:

- **H1.** Participants play better right after watching the expert than right after watching the novice.

  - Estimand: mean payoff and mean entropy of the play games that follow an expert demonstration minus those that follow a novice demonstration, each averaged over both rounds and both social conditions (four cells against four).
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign payoff $+$, entropy $-$; Holm over the two outcomes.
  - Contrast: `contrast(emm, list(w = m(demo == "expert") - m(demo == "novice")))`
- **H2.** Participants play better after watching the novice demonstrator the second time, in round 2, than after watching the novice the first time, in round 1.
  - Estimand: mean payoff and mean entropy of the after-novice play games in round 2 minus the same games in round 1, averaged over the two social conditions.
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign payoff $+$, entropy $-$; Holm over the two outcomes.
  - Contrast: `contrast(emm, list(w = m(demo == "novice" & round == 2) - m(demo == "novice" & round == 1)))`
- **H3.** In round 2 participants concentrate their choices on fewer arms than in round 1.
  - Estimand: mean entropy of the six choices of a play game in round 2 minus round 1, averaged over all play games of both social conditions.
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign entropy $-$; one outcome, no correction.
  - Contrast: `contrast(emm, list(w = m(demo != "none" & round == 2) - m(demo != "none" & round == 1)))`
- **H4.** After watching the expert and novice demonstrators each in both rounds 1 and 2, participants choose the expert more often than the novice.
  - Estimand: the probability that a participant chooses the expert in the choice game that ends a social session, estimated with a participant-level intercept; the claim is that it exceeds one half.
  - Model: the mixed-effects model of partner choice, Section 4.2.
  - Rule: Wald z-test on β_0, two-sided $\alpha = 0.05$ with 95% CI; predicted sign $\beta_0 > 0$ on the logit scale.
  - Fit: `glmer(chose_expert ~ condition + (1 | participant), family = binomial)`, sum-to-zero coding; the intercept is the group-level log-odds of choosing the expert, averaged over the two conditions.
- **H5.** Participants play better after watching the expert, in both rounds, than solo participants do in their final game.
  - Estimand: mean payoff of the play games that follow an expert

demonstration minus mean payoff of the solo condition's fourth and final play game of the round, averaged over rounds. The solo game comes after 18 own choices; the after-expert game after 6 (expert-first) or 18 (novice-first) revealed choices, so solo has at least as much information in every comparison.
  - Model: the mixed-effects model of play outcomes, Section 4.1.
  - Rule: Satterthwaite t-test on the contrast, two-sided $\alpha = 0.05$ with 95% CI; predicted sign payoff +; one outcome, no correction.
  - Contrast: `contrast(emm, list(w = m(demo == "expert") - m(demo == "none" & play == "second")))`

Secondary Claims:

- **S1.** Arm choices are predicted better by models that use the estimate of the expert's beliefs than by models that use the estimate of the most recently watched demonstrator's beliefs (S1a) or of the novice's beliefs (S1b).
  - Estimand: out-of-sample predictive accuracy, ΔELPD, of the full policy model reading the expert's belief minus the same model reading the last-watched demonstrator's belief (S1a) and minus the same model reading the novice's belief (S1b), with the standard error paired over participants.
  - Models: ROS of expert vs recent (S1a) and vs novice (S1b).
  - Rule: not a test; each part ΔELPD, Equation 7, > 2 paired SE, reported with its SE.
- **S2.** Arm choices are predicted better by models that use the estimate of the expert's beliefs than by models that simply count the successes and failures of 1. the most recent demonstrator they watched (S2a), 2. the expert demonstrator they watched (S2b), or 3. every demonstrator they watched (S2c).
  - Estimand: ΔELPD of the full model minus the same model with the expert's belief replaced by a Beta count of the visible choices of the last demonstration (S2a), of the expert's demonstration on this bandit (S2b), and of every choice seen on this bandit this round (S2c).
  - Models: ROS of expert vs naive-recent (S2a), vs naive-expert (S2b) and vs naive (S2c).
  - Rule: not a test; each part ΔELPD > 2 paired SE, reported with its SE.
- **S3.** Arm choices are predicted better by models that use the estimate of the expert's beliefs alone than by models that use only own experience (S3a). Similarly, models that use the estimate of the expert's beliefs as well as own experience predict choices better than models that use only own experience (S3b).
  - Estimand: ΔELPD of the model with the expert's belief and no own outcomes minus the model with own outcomes and no belief (S3a), and of the full model minus the model with own outcomes and no

belief (S3b).
  - Models: RS vs RO (S3a), ROS vs RO (S3b).
  - Rule: not a test; each ΔELPD > 2 paired SE, reported with its SE.
- **S4.** Participants are more influenced by the expert's belief in round 2 after watching the novice than in round 1 after watching the novice.
  - Estimand: in the switch model, the group-level change from round 1 to round 2 in the weight placed on the expert's belief on play games that follow the novice, on the logit scale, $\Delta = \delta_N$.
  - Model: Switch-by-round, Eq. 6.
  - Rule: posterior interval, not a test; 95% ETI of Δ excludes 0 on the predicted side.

Rules and verdicts. The Holm correction applies across the outcomes within a hypothesis. Verdicts are *supported*, *not supported*, *contradicted* or, for the secondary claims, *not evaluable* (Section 4.4). No smallest effect of interest is registered.

4 Models and analysis

4.1 Mixed-effects model of play outcomes

Model. One linear mixed model per outcome (payoff, reward, entropy), fitted with `lmerTest::lmer` on one row per play game (`src/r/descriptive.R`):

```
outcome ~ condition * play * round + (1 | participant)
```

- `condition`: factor with three levels: expert-first, novice-first and solo; group-level.
- `play`: factor with two levels, referring to the 1st and 2nd play games in the social conditions, and 2nd and 4th play games of the solo condition; group-level.
- `round`: factor with two levels, 1 and 2, each on a new bandit; group-level.
- `condition * play * round`: the three main effects and all their interactions, under sum-to-zero coding; all group-level.
- `(1 | participant)`: a random intercept per participant.

In the solo condition the 2nd and 4th play games are used to match the social condition's 1st and 2nd play games. This choice was made in order to have the same number of trials of observations before these play games in both the social and solo conditions.

Contrasts. The model results in 12 cell means per outcome type (payoff, reward, entropy), one per combination of condition, play and round. Each cell is read off the fit with `emm <- emmeans(fit, ~ condition * play * round)` with Satterthwaite degrees of freedom. Each hypothesis is the mean of one set of cells minus the mean of another, computed as `contrast(emm, list(w = m(A) - m(B)))` with `A` and `B` for each claim given in Section 3.

Reward. In addition to payoff and entropy we also report observed reward with its 95% confidence interval, for the same contrasts of payoff (H1, H2 and H5). It reflects the same quantity but noisier.

4.2 Mixed-effects model of partner choice

Model. A single logistic mixed model fitted with `lme4::glmer` on one row per social condition per participant (the final choice game in each, so 2 rows total per participant), with `chose_expert` equal to 1 if the participant chose the expert (`src/r/partner_choice.R`):

```
chose_expert ~ condition + (1 | participant), family = binomial
```

- `condition`: factor with two levels, expert-first and novice-first, under sum-to-zero coding; group-level. In expert-first the novice was watched last, in novice-first the expert.
- the intercept: the group-level log-odds that the expert is chosen, averaged over the two conditions; H4's estimand.
- `condition`'s coefficient: half the expert-first minus novice-first difference in log-odds; a difference between the conditions is a lean toward the last-watched demonstrator.
- `(1 | participant)`: a random intercept per participant, a normal deviation of each participant's own log-odds; the only per-participant term.
- `family = binomial`: the outcome is 0/1, modelled as Bernoulli with its probability on a log-odds scale.

Reporting. The intercept and its 95% confidence interval are reported on the logit scale and the inverse logit (probability scale). The condition coefficient is reported with its 95% confidence interval as an exploratory check of a difference between the conditions; it carries no rule. In addition the mean raw rate of expert picks is reported with a 95% Wilson score interval, for all sessions and for each condition.

4.3 Policy models

Setup. A policy model is a generative model of a participant's choices during the play games. The first part of the pipeline uses a transformer to generate predictions of each demonstrator's beliefs over the payoffs of the arms of the bandit, given each of the demonstrators observed choices and outcomes. These are called the readouts. The second part of the pipeline determines the different terms that enter the policy model, including the demonstrators' beliefs, the participant's own outcomes, and the participant's own previous step's choice and reward outcome. Each of these terms are combined into what becomes the policy model, and is fitted to the participant's choices and outputs a log-likelihood of the choices given the model computed as a soft-max.

The policy models are fitted to every shot of the play games of the two social conditions (expert-first and novice-first). For each of the two conditions there

are two rounds, two play games and six shots, making 48 shots total per participant. Each shot is a trial t of participant i choosing arm a_t from $K = 12$ arms. The readouts from the transformer are the parameters of a Beta distribution per arm k, $(\alpha_k^{\text{demo}}, \beta_k^{\text{demo}})$. The participant's own outcomes are the counts of successes and failures on each arm k in the current play game, s_k^{curr} and f_k^{curr}, and in the earlier play game of the same round, s_k^{prev} and f_k^{prev}. The previous arm chosen is a_{t-1} with its reward outcome $r_{t-1} \in \{0, 1\}$.

The full model, ROS. The full model includes all terms: the repeat term (R), a bias toward repeating the previous arm scaled by whether it just won, the own-experience term (O) and the social-belief term built on the readouts (S), hence the name ROS. Each term carries its own weight.

$$\alpha_k = \kappa_i\, \alpha_k^{\text{demo}} + \lambda_{\text{w}}\, s_k^{\text{curr}} + \lambda_{\text{c}}\, s_k^{\text{prev}}, \qquad \beta_k = \kappa_i\, \beta_k^{\text{demo}} + \lambda_{\text{w}}\, f_k^{\text{curr}} + \lambda_{\text{c}}\, f_k^{\text{prev}}, \tag{1}$$

Each arm's belief is a Beta(α_k, β_k) distribution. Under Thompson sampling an arm is chosen with the probability that its draw from this distribution is the highest among the twelve arms. This is

$$p_k = \int_0^1 f_k(x) \prod_{j \neq k} F_j(x)\, dx, \tag{2}$$

with f_k and F_k the probability density function (pdf) and cumulative distribution function (CDF) of Beta(α_k, β_k), respectively. The log of p_k is the logit of arm k. A win-stay lose-shift tilt adds a bonus, with sign depending on the last reward, to the logit of the previous arm,

$$\ell_k = \log p_k + (\rho_{0,i} + \rho_{\text{win},i}\, r_{t-1})\, \mathbb{1}[k = a_{t-1}], \tag{3}$$

The final choice is a softmax over the logits, at unit temperature, with a fixed lapse,

$$\pi_k = \frac{\exp \ell_k}{\sum_j \exp \ell_j}, \qquad P(a_t = k) = (1 - \varepsilon)\, \pi_k + \frac{\varepsilon}{K}, \qquad \varepsilon = 0.02. \tag{4}$$

Without the readout, the readout term of Equation 1 is replaced by 1, so $\alpha_k = 1 + \lambda_{\text{w}}\, s_k^{\text{curr}} + \lambda_{\text{c}}\, s_k^{\text{prev}}$ and likewise β_k with the failure counts, or $\alpha_k = \beta_k = 1$ for the repeat term alone. Without the win-stay lose-shift $\ell_k = \log p_k$ and $\pi_k = p_k$.

Hierarchical and group-level parameters. κ_i and ρ_i are fitted per participant since every trial carries evidence for them: κ_i scales the readout, and is present on all 48 shots, and ρ_i acts on the previous arm, which exists on every trial after the first. However λ_{w} and λ_{c} only act on the arms a participant

has already attempted. Because there are 12 arms, 48 trials total results in a participant providing very few informative shots to estimate them. Hence they are pooled across participants. The lapse rate ε is fixed at 0.02 rather than fitted. This allows every arm to have some probability of occurring so that a single stray trial does not dominate the likelihood.

The ladder. Every other model is Equation 1 to Equation 4 with a term replaced or removed. ROS is the model with all terms. Models missing the repeat term drop the R, and similarly for O and S. The models of S1 to S3, each described as a change from ROS:

Model	Terms included
ROS	readout of the expert (S), own counts (O), repeat tilt (R)
recent	same as ROS but with the readout of the last-watched demonstrator
novice	same as ROS but with the readout of the novice
naive-recent	same as ROS but with $\text{Beta}(1,1) + \kappa_i\times$ the visible shots of the last demonstration in place of the readout
naive-expert	same as ROS but with $\text{Beta}(1,1) + \kappa_i\times$ the visible shots of the expert's demonstration on current bandit in place of the readout
naive	same as ROS but with $\text{Beta}(1,1) + \kappa_i\times$ every shot seen on the bandit in current round including play and watch games
RS	same as ROS without the own counts
RO	same as ROS without the readout

Random is uniform choice with nothing fitted and included as a floor.

The Switch models. Two ROS policies sharing the same parameters for the repeat, own-experience and social terms (ρ, λ and κ_i), one using the expert's readout and the other the novice's, are mixed,

$$P(a_t = k) = (1-\varepsilon)\left[\omega_t\,\pi_k^E + (1-\omega_t)\,\pi_k^N\right] + \frac{\varepsilon}{K}, \tag{5}$$

where ω_t is the participant's weight on the expert's policy for that play game, $\omega_{E,i}$ when the game follows the expert and $\omega_{N,i}$ when it follows the novice. Each is a participant-level weight and is hierarchical on the logit scale (e.g. $\text{logit}\,\omega_{E,i} = \mu_E + \tau_E z_{E,i}$). Switch-by-round lets each weight change based on the round at the group level,

$$\begin{aligned} \text{logit}\,\omega_{N,i,r} &= \mu_N + \tau_N z_{N,i} + \delta_N x_r, \qquad x_1 = -\tfrac{1}{2},\ x_2 = +\tfrac{1}{2}, \\ \Delta = \delta_N &= \text{logit}\,\omega_N(2) - \text{logit}\,\omega_N(1), \end{aligned} \tag{6}$$

and likewise for E with δ_E.

Model	Terms included
Switch	two ROS policies, one on the expert's readout and one on the novice's, mixed by Equation 5
Switch-by-round	same as Switch with the round term Equation 6

Priors. Hierarchical parameters are non-centred. The group mean is μ, the group spread is τ and the standard-normal participant deviation is z_i. They are combined as $\mu + \tau z_i$ on the scale named in the table below. Parameters with the participant subscript i are hierarchical. The rest are group-level. The exploratory variants add parameters whose priors are in the fitting code at the registered tag.

Parameter	Scale	Prior
κ_i	log	$\mu_\kappa \sim \mathcal{N}(0, 0.5)$, $\tau_\kappa \sim$ HalfNormal(0.3)
λ_{w}	log	$\mathcal{N}(0, 1.5)$, group level
λ_{c}	linear	HalfNormal(4), group level; fixed at 0 in ROS-reset
$\rho_{0,i}$	linear	$\mu \sim \mathcal{N}(0, 2)$, $\tau \sim$ HalfNormal(1)
$\rho_{\mathrm{win},i}$	linear	$\mu \sim \mathcal{N}(0, 2)$, $\tau \sim$ HalfNormal(1)
ε	linear	fixed at 0.02
$\omega_{E,i}$, $\omega_{N,i}$	logit	$\mu \sim \mathcal{N}(0, 1.5)$, $\tau \sim$ HalfNormal(1)
δ_E, δ_N	logit	$\mathcal{N}(0, 1)$, group level

4.4 Fitting and model comparison

Thompson probabilities. Equation 2 is evaluated with trapezoidal quadrature on a uniform grid of the logit scale, over a dense grid of κ, λ_{w} and λ_{c} and interpolated. This was checked to be accurate to about 10^{-7}.

Sampling and gate. Every model is fitted with NUTS in NumPyro. Here we use four chains of 1,000 warm-up iterations and 1,000 sampling iterations with the sampler set to target acceptance of 0.9. A fit passes when $\widehat{R} < 1.01$ and bulk and tail effective sample sizes are at least 400 on every parameter and no divergent transitions. A failing fit is refitted at 2,000 + 2,000 iterations with target acceptance increased to 0.95, then 4,000 + 4,000 with target acceptance increased to 0.99. Any fit that fails the last rung is not reported.

Comparison. Models are compared by expected log predictive density (ELPD) estimated with Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) from the pointwise log-likelihoods of the posterior draws, summed over trials. For any two given models A and B, where d_i is the sum over participant i's shots of the pointwise difference between the two models, and N is the number of participants we compute

$$\Delta\text{ELPD}_{AB} = \sum_{i=1}^{N} d_i, \qquad \text{SE}_{AB} = \sqrt{N}\ \text{sd}(d_1, \ldots, d_N), \tag{7}$$

so the standard error is paired over participants. A comparison supports A when $\Delta\text{ELPD}_{AB} > 2\,\text{SE}_{AB}$. A trial whose Pareto k exceeds 0.7 is an unreliable importance estimate. Every session containing such a trial is held out entirely and the model is refitted without it, and all of that session's trials enter by their true leave-one-session-out log predictive density under that refit. The number of sessions refitted per model is reported beside each ΔELPD.

Verdicts. A comparison is contradicted when the rival is favoured by more than two standard errors. A claim is not evaluable when a fit it needs failed the gate; the comparison is still reported.

Recovery checks. Two simulation checks are run and reported beside the claims. (a) Parameter recovery of δ_N: 25 datasets of 100 simulated participants are generated from posterior draws of the Switch-by-round fit to the analysed data, each played through the full experiment with new transformer Beta predictions before every play game, and refitted with the registered pipeline. We report how many of the 25 intervals cover the true δ_N, how many exclude zero on the same side as the truth, and the bias, the mean of the posterior means minus the truths. (b) Model recovery for S1, S2 and S4. For S1 and S2 the pair is the full model against its nearest rival, the model with the smallest ΔELPD from ROS within that claim in the fit to the analysed data, after the Pareto-k fallback refits of Section 4.4 (on the pilot, recent for S1 and naive for S2); any of the S1 or S2 rivals can be the pair's second model. For S4 the pair is Switch-by-round against Switch. Five datasets are generated from each model of the pair and both are fitted to each. We report how often each generating model wins its own datasets and the other model's.

4.5 Posterior predictive checks

Free-running check. The fitted ROS and Switch-by-round generative models are each simulated to play the experiment and their behaviour is compared with the participants'. At each of 50 posterior draws spaced evenly over the chains, every participant replays their own two social sessions with their drawn parameters. The fitted policy takes every shot of every play game, the demonstrations are the same exact ones the participant observed in the watch games, and the transformer makes new Beta predictions before each play game so that the read-out entering a round's second play game reflects the first simulated play. Five statistics are computed on each replay exactly as on the human data, each a mean over participants of a per-participant mean:

1. the after-novice payoff rise from round 1 to round 2 (H2's quantity)
2. the entropy fall from round 1 to round 2 over the social play games (H3's quantity)

3. payoff after the expert minus after the novice within novice-first sessions
4. the share of shots that repeat the previous arm after a win
5. the same share after a loss

For each statistic we report the human value against the 95% ETI of the statistic over the 50 replays, for every statistic and both models.

Split-half generative check (exploratory). The same two models are fitted on a random half of the participants (fixed seed) and the other half is replayed at each of the 50 draws with parameters drawn fresh from the fitted group-level hierarchy rather than their own posteriors, so the interval shows what the population model predicts for participants it has never seen. The five statistics of the held-out half are reported against it in the same way, beside the main check.

5 Exploratory analyses

The analyses below are ones that may be run, not a commitment to run every one of them. None of them carries a decision rule: each is reported with its estimate and 95% interval.

	Analysis	What is reported	Expectation
E1	after-expert round change	round 2 minus round 1 in the after-expert cells, from the model of Section 4.1, all outcomes	no fall; may rise
E2	the other social cells vs solo's final game	H5's comparison with the after-novice cells (E2a) and with the social cells averaged over demonstrators (E2b) in place of the after-expert cells, and H5 within round 2 alone (E2c) (payoff)	after-novice near zero; the average positive and smaller than H5; round 2 like H5
E3	the after-novice improvement vs solo practice	H2's difference minus the same round difference in solo's first and second play, from the model of Section 4.1, all outcomes; the solo term alone is reported as E3s	payoff positive; entropy ambiguous, since solo entropy also falls with practice
E4	reward	the reward version of every contrast above, H1, H2, H5 and E1 to E3, with its 95% interval (Section 4.1)	same sign as payoff, wider intervals

	Analysis	What is reported	Expectation
E5	condition breakdown of the after-novice round change	H2's contrast within expert-first and within novice-first separately; in novice-first the round 2 after-novice game is on a new bandit before the expert has been seen on it, so any improvement there is carried over from round 1	none stated
E6	sensitivity of H1, H2, H3 and H5	the model of Section 4.1 refitted two ways, each reporting estimate, interval and uncorrected p-value without Holm or verdict: (a) random slopes for round and play by participant, coded $-\frac{1}{2}$ and $+\frac{1}{2}$, reduced to uncorrelated slopes and then to the intercept-only model if singular or non-convergent, the specification reached and the slope standard deviations of every specification tried reported, so that a slope variance at zero is visible rather than discarded; (b) session order as a numeric covariate in the fixed part, cell means taken at its mean	same verdicts
E7	reported ladder rungs	paired ΔELPD with its SE, Equation 7, as rows of the choice stage: the ladder's difference from ROS for OS (repeat), ROS+persev (perseveration) and ROS-reset (own outcomes not carried between play games), and the two paired rows ROS minus RS (own experience on top of the readout) and S minus O (social belief against own experience)	repeat positive; S over O positive; the rest near zero

	Analysis	What is reported	Expectation
E8	δ_E	the round change of the weight on the expert's belief after the expert, Equation 6, as its posterior mean and 95% ETI from the Switch-by-round fit	no fall
E9	Switch-by-round vs Switch	ΔELPD of the round term; beside it, model recovery for this pair run as in check (b) of Section 4.4	positive
E10	robustness of S4	three refits, each reporting posterior means with 95% ETIs: Switch-by-round with a group-level round trend on $\log \kappa$ (Switch-by-round+ktrend: δ_κ and δ_N); ROS with the lapse rate free at group level (ROS+eps: ε and the group-level κ); Switch-by-round with the lapse rate free (Switch-by-round+eps: ε and δ_N)	Δ unchanged in sign and interval
E11	the unseen-expert readout	the five analyses (a) to (e) of the table below	the expert's readout matters before the expert is seen on the bandit, in round 2
E12	recency in the partner choice	the condition term of the model of Section 4.2, under sum-to-zero coding half the expert-first minus novice-first difference on the logit scale; in expert-first the novice was watched last, in novice-first the expert, so a term near zero means the choice tracks expertise rather than recency	near zero

	Analysis	What is reported	Expectation
E13	practice within a round	the solo condition's second minus first play game, from the model of Section 4.1, all outcomes: the effect of extra play on the same bandit without a demonstration in between	payoff near zero or a small rise; entropy a small fall

6 Sampling plan

6.1 Pilot Data

Multiple pilots of this task were run prior to this preregistration. These were used to develop the task along with the specific framing, reward labeling and clarity of partners and zones (the bandits) to clearly distinguish them from each other. These were also helpful as data for developing and debugging the models and finally in generating the hypotheses and estimating the sample size. Only the final pilot, referred to as pilot E, was used in the power analysis since only that pilot used the finalized version of the task. Pilot E was run with 38 participants after all exclusions.

6.2 Power analysis and sample size

Method. Power is estimated only for the primary confirmatory hypotheses H1 to H5. Pilot E's 38 participants were resampled with replacement to generate 500 different simulated datasets at the following sample sizes: $N \in \{200, 250, 300, 350, 400, 500, 600, 700, 800\}$. Every contrast is evaluated on each simulated dataset using the same models in Section 4.1 with Holm correction applied within each hypothesis. The proportion of simulations that support each hypothesis is reported as the power estimate. We also report the power for each hypothesis at half the effect size observed in pilot E. The script is `src/r/power_analysis.R`.

Decision rates at the pilot effect.

Test	200	250	300	350	400	500	600	700	800
H1, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H1, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H2, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H2, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H3, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H4, choice	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H5, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Decision rates at half the pilot effect:

Test	200	250	300	350	400	500	600	700	800
H1, payoff	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H1, entropy	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H2, payoff	.67	.72	.86	.91	.93	.96	.98	1.00	1.00
H2, entropy	.98	.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H3, entropy	.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H4, choice	.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H5, payoff	.95	.97	.99	1.00	1.00	1.00	1.00	1.00	1.00

Secondary claims. None of the secondary claims are powered, and each is reported with its paired standard error or posterior interval.

Chosen sample size. We plan to recruit $N = 350$ participants who survive the exclusion rules of Section 6.4. At $N = 350$ the lowest-powered claim, H2 on payoff, is at 0.91 at half the pilot effect.

6.3 Participants and sampling

Recruitment. Participants are recruited on Prolific. The following prescreens are applied: resident in the United States, aged 18 to 50, fluent in English, and no colour-vision deficiency. These are all self-reported on Prolific and unverified. In addition anyone who took prior pilots of this task or related experiments of our earlier work is excluded in advance through Prolific's prescreening feature.

Stopping rule. Recruitment will run in batches on Prolific until we have N=350 participants that pass all exclusion rules. After each batch only the exclusion script is run to count survivors. Batches are sized so that the last batch is expected to reach N with a small overshoot. Every participant in the final batch who passes the rules will be included in the analysis.

Payment. A base payment of $2 will be paid to all participants who complete the task within the 60 minute cutoff. $2 assumes a payment of $12 per hour at a 10 minute completion pace. Pilot E median time to completion was 12 minutes. In addition participants will receive a 2 cent bonus for every point earned during the game. Participants who decline consent are returned right away and unpaid.

Consent and data sharing. Participants consent on the first page of the experiment. The single consent page discloses that the data will be de-identified, analysed and shared publicly, and agreeing to take part is consent to all of this. Every analysed participant's anonymised trial data are released (Section 7). Age and sex, from Prolific's profile export, are reported only as summaries over the analysed sample and are not released per participant.

6.4 Exclusions

Every exclusion rule lives in `src/r/preprocess_participant_data.R`, which writes a ledger of counts per rule (`exclusions.csv`). Nothing else is excluded.

	Level	Rule	Reason	Pilot E
X1	participant	returned the submission on Prolific	withdrawal	8 of 49
X2	participant	answered yes to Prolific's colour-blindness prescreen	game asteroids and rewards and partners are in various colours and otherwise may be hard to tell apart	2
X3	participant	completion in under 300 s	300 s is roughly the time it takes to click through the game as fast as possible; indicative of not paying attention	0
X4	participant	completion takes over 60 min	study left unattended	1
X5	session	no watch or play shot logged	no data	0
X6	session	less than 2 rounds or less than 4 games	incomplete session	0
X7	session	number of shots not exactly 6	incomplete or corrupted game	0
X8	session	page reloaded during game	a reload restarts the game giving them extra shots to learn from	0
X9	participant	more than one page load of the study	a reload restarts the study	0
X10	participant	no change to starting position	participant never moved	0
X11	participant	not exactly three sessions after exclusions	dropouts, missing or repeated sessions	0

A Prolific time-out is not considered an exclusion and participant data will still be included if completed and within the 60-minute window (X4). This is measured on the game's own log, not by Prolific. Such participants are still paid.

7 Reporting and data availability

Reporting. Once data collection for the full study is complete, the results are written by the pipeline's report stage into `results.md` in the order of the pipeline's manifest (`numbers.py`). That file is released with the data.

Data availability. Released publicly with the results: `preprocessed_participant_data_public.csv`, the analysed participants' anonymised trial data (the preprocessed file without timestamps); `exclusions.csv`, the ledger of counts per exclusion rule; `demographics_summary.csv`, age and sex summarised over the analysed sample; and the aggregate outputs of every analysis stage. No per-participant demographics are released. From the public file, the tagged code and the checkpoint named above, inference and every analysis stage from `games` onward can be rerun to regenerate `results.md`.